Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

Options and Futures

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Introduction
▷ Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

The Binomial Model and the FTAP

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Key concepts

Introduction
Binomial and FTAP
▷ Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- arbitrage (and absence of arbitrage)
- no-arbitrage pricing
- replicating portfolio
- hedging
- state prices
- risk-neutral probability
- risk-neutral valuation

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Derivatives pricing in the static binomial model

Introduction
Binomial and FTAP
▷ Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- One-period model (say time interval of length T).
- A stock initially trades at S_0. Its price can move up to uS_0 or down to dS_0, where $0 < d < 1 < u$. The probability of the stock price going up is p.
  - For short, we will refer to the two states of the world as states u and d.
- In addition, there is a money market account (MMA) which allows riskfree borrowing and lending at a continuously compounded interest rate r, with $d < e^{rT} < u$.
- Consider a derivative with end-of-period payoff V_u or V_d contingent on the realization of the end-of-period stock price.
- **Problem:** Given $S_0, u, d, p, r, T, V_u$and V_d, what should the initial price V_0 of the derivative be?

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Examples

Introduction
Binomial and FTAP
▷ Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Forward contract with forward price F:
  - $V_u = uS_0 - F$
  - $V_d = dS_0 - F$
- European call option with strike price K:
  - $V_u = (uS_0 - K)^+$
  - $V_d = (dS_0 - K)^+$
- European put option with strike price K:
  - $V_u = (K - uS_0)^+$
  - $V_d = (K - dS_0)^+$

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Replicating portfolio

Introduction
Binomial and FTAP
Binomial Model
▷ Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Buy δ units of the stock and invest an amount β in the MMA (negative β means borrowing) so as to replicate the payoff of the derivative.
- Need to find δ and β such that

$$\delta U S_0 + \beta e^{rT} = V_u,$$
$$\delta d S_0 + \beta e^{rT} = V_d$$

- Unique solution:

$$\delta = \frac{V_u - V_d}{(u-d)S_0}, \quad \beta = e^{-rT} \cdot \frac{uV_d - dV_u}{(u-d)}.$$

- Cost of the replicating portfolio:

$$\delta S_0 + \beta = e^{-rT}[qV_u + (1-q)V_d],$$

where

$$q := \frac{e^{rT} - d}{u - d}.$$

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No-arbitrage pricing

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
▷ No-arbitrage pricing
Risk-neutral valuation
FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- No-arbitrage argument: Since the derivative and the replicating portfolio have the same contingent payoffs, the price of the derivative V_0 must be equal to the cost of the replicating portfolio, i.e.

$$V_0 = \delta S_0 + \beta = e^{-rT}[qV_u + (1-q)V_d].$$

- Otherwise there would be an arbitrage opportunity:
  - If $V_0 < \delta S_0 + \beta$, buy the derivative and sell the replicating portfolio;
  - If $V_0 > \delta S_0 + \beta$, sell the derivative and buy the replicating portfolio.

  In both cases, you earn money initially and break even in any future state of the world.

Alternative no-arbitrage argument (delta-hedging)

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
▷ No-arbitrage pricing
Risk-neutral valuation
FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Construct a zero-value portfolio as follows:
  - Sell one unit of the derivative,
  - Buy δ shares of the stock,
  - Invest the proceeds in the MMA: $\beta = V_0 - \delta S_0$.
- Payoff in state u: $-V_u + \delta u S_0 + e^{rT}(V_0 - \delta S_0)$.
- Payoff in state d: $-V_d + \delta d S_0 + e^{rT}(V_0 - \delta S_0)$.
- Delta-hedging: Choose δ so that the payoff is riskless (i.e. the same in both states)

$$-V_u + \delta u S_0 = -V_d + \delta d S_0$$
$$\Rightarrow \quad \delta = \frac{V_u - V_d}{(u-d)S_0}.$$

- This is the same δ as in the replicating portfolio for the derivative.

No-arbitrage pricing

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
▷ No-arbitrage pricing
Risk-neutral valuation
FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Since the delta-hedged portfolio costs nothing and has a riskless payoff, this payoff must in fact be zero. Otherwise there is an arbitrage opportunity:
  - If the payoff is positive, go long in this portfolio;
  - If the payoff is negative, go short.
- Thus, we must have:

$$-V_u + \delta u S_0 + e^{rT}(V_0 - \delta S_0) = 0.$$

- Substituting for δ and solving for V_0, we end up with the same result as before, i.e.

$$V_0 = e^{-rT}[qV_u + (1-q)V_d],$$

where

$$q := \frac{e^{rT} - d}{u - d}.$$

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Risk-neutral valuation

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
▷ Risk-neutral valuation
FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Observation: The probability p of the price of the underlying going up does not enter the expression for V_0, i.e. the price of the derivative does not depend on the expected change in the stock price, or on the expected payoff of the derivative!
- However, if we interpret q as the probability of state u and let Q denote the corresponding probability measure, our expression for the derivative price is equal to the expected payoff of the derivative under Q, discounted at the riskfree rate r:

$$V_0 = e^{-rT}\mathbb{E}^Q(V_T)$$

- Essence of risk-neutral valuation: Absence of arbitrage implies the existence of a probability measure Q under which securities are priced as if investors were risk-neutral.
  - If investors were actually risk-neutral, we would have

$$V_0 = e^{-rT}\mathbb{E}(V_T).$$

Risk-neutral valuation (cont'd)

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
▷ Risk-neutral valuation
FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- The same principle applies to the underlying as well.
- Under Q, the expected value of S_T is

$$\begin{aligned}\mathbb{E}^Q(S_T) &= quS_0 + (1-q)dS_0 \\ &= e^{rT}S_0,\end{aligned}$$

- i.e. the expected return on the stock under Q is equal to the riskfree rate. Or, equivalently,

$$S_0 = e^{-rT}\mathbb{E}^Q(S_T).$$

  i.e. the price of the stock is equal to the expected payoff of the stock under Q, discounted at the riskfree rate.
- Q is referred to as the risk-neutral probability measure.
- Note that $0 < q < 1$ if and only if $d < e^{rT} < u$.

The Fundamental Theorem of Asset Pricing

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- S states of the world: $1, 2, \ldots, S$.
- N securities: $1, 2, \ldots, N$.
- The state-contingent payoffs of asset n are given by a vector in $\mathbb{R}^S$:

$$\mathbf{d_n} = \begin{bmatrix} d_{1n} \\ \vdots \\ d_{Sn} \end{bmatrix}$$

- The N assets taken together can be represented by the $S \times N$ matrix $\mathbf{D}$:

$$\mathbf{D} = [\mathbf{d_1} \ldots \mathbf{d_N}] = \begin{bmatrix} d_{11} & \cdots & d_{1N} \\ \vdots & \ddots & \vdots \\ d_{S1} & \cdots & d_{SN} \end{bmatrix}$$

- Security prices are given by the column vector $\boldsymbol{\pi} \in \mathbb{R}^N$.

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The Fundamental Theorem of Asset Pricing

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

$$
\begin{aligned}
S &= 2, \\
N &= 2, \\
\mathbf{D} &= \begin{pmatrix} uS_0 & 1 \\ dS_0 & 1 \end{pmatrix} \\
\boldsymbol{\pi} &= \begin{pmatrix} S_0 \\ e^{-rT} \end{pmatrix}
\end{aligned}
$$

Note that the MMA can be thought of as a riskless bond.

Portfolios

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- A portfolio can be described by a vector of holdings $\boldsymbol{\theta} \in \mathbb{R}^N$.
- The cost of portfolio $\boldsymbol{\theta}$ (which can be positive or negative) is $\sum_{n=1}^{N} \pi_n \theta_n$, which we can write as[1] $\boldsymbol{\pi}^\top \boldsymbol{\theta}$ or $\boldsymbol{\pi} \cdot \boldsymbol{\theta}$
- The payoff of portfolio $\boldsymbol{\theta}$ in state s is $\sum_{n=1}^{N} \theta_n d_{sn}$.
- The state-contingent payoff of portfolio $\boldsymbol{\theta}$ is given by $\mathbf{D}\boldsymbol{\theta} \in \mathbb{R}^S$.
- The "dividend stream" resulting from portfolio θ (including the payoff at the initial date) is:

$$\begin{bmatrix} -\boldsymbol{\pi}^\top \\ \mathbf{D} \end{bmatrix} \boldsymbol{\theta} \quad \in \mathbb{R}^{S+1}.$$

[1] The symbol $^\top$ denotes "transpose"

Arbitrage

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Notation: When dealing with a vector $\mathbf{x}$, we write
  - $\mathbf{x} \geq \mathbf{0}$ if all components are ≥ 0,
  - $\mathbf{x} > \mathbf{0}$ if all components are ≥ 0 and at least one is > 0,
  - $\mathbf{x} \gg \mathbf{0}$ if all components are > 0.
- A portfolio $\boldsymbol{\theta} \in \mathbb{R}^N$ is an arbitrage if

$$\begin{bmatrix} -\boldsymbol{\pi}^\top \\ \mathbf{D} \end{bmatrix} \boldsymbol{\theta} > 0.$$

- Interpretation: An arbitrage allows you to make money at no cost and with no risk of loss.

FTAP

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- **Proposition:** $(\mathbf{D}, \boldsymbol{\pi})$ does not admit arbitrage if and only if there exists a strictly positive vector $\boldsymbol{\psi} \gg 0$ in $\mathbb{R}^S$ such that

$$\boldsymbol{\pi}^\top = \boldsymbol{\psi}^\top \mathbf{D},$$

i.e. for every asset n,

$$\pi_n = \boldsymbol{\psi} \cdot \mathbf{d_n} = \sum_{s=1}^{S} \psi_s d_{sn}.$$

- The positive numbers $\psi_1, \ldots, \psi_S$ are called state prices.
- The existence of state prices is necessary and sufficient for all securities to be priced consistently and no free lunch to be available.
- A security paying 1 in state s and zero otherwise is known as an Arrow security. The state price ψ_s is the price of the Arrow security corresponding to state s.

Sketch of the proof of the FTAP

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

Part 1: If state prices exist, then there is no arbitrage.

- Given a state price vector ψ,

$$\begin{bmatrix} 1 & \boldsymbol{\psi}^\top \end{bmatrix} \begin{bmatrix} -\boldsymbol{\pi}^\top \\ \mathbf{D} \end{bmatrix} \boldsymbol{\theta} = (-\boldsymbol{\pi}^\top + \boldsymbol{\psi}^\top \mathbf{D})\boldsymbol{\theta} = 0,$$

  for all portfolios θ.

- If θ were an arbitrage,

$$\begin{bmatrix} -\boldsymbol{\pi}^\top \\ \mathbf{D} \end{bmatrix} \boldsymbol{\theta} > 0,$$

  so that

$$\begin{bmatrix} 1 & \boldsymbol{\psi}^\top \end{bmatrix} \begin{bmatrix} -\boldsymbol{\pi}^\top \\ \mathbf{D} \end{bmatrix} \boldsymbol{\theta} > 0,$$

  a contradiction.

Proof of the FTAP (cont'd)

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

Part 2: If there is no arbitrage, then state prices exist. The proof is more involved.

- The law of one price (LOP) is a necessary condition to prevent arbitrage opportunities. Hence, there must exist a linear pricing function, i.e. a vector of state prices.
- Absence of arbitrage is stronger than the LOP and requires strict positivity of state prices (if $\psi_s \leq 0$, simply holding the Arrow security paying in state s would constitute an arbitrage).

No-arbitrage pricing using the FTAP

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- A payoff (or claim or asset) $\mathbf{d} \in \mathbb{R}^S$ is attainable if it can replicated by the existing assets, i.e. if there exists a replicating portfolio $\boldsymbol{\theta}_d$ such that $\mathbf{D}\boldsymbol{\theta}_d = \mathbf{d}$.
- The no-arbitrage price of an attainable payoff $\mathbf{d}, \pi_d$, must be equal to the cost of the replicating portfolio, i.e. $\pi_d = \boldsymbol{\pi} \cdot \boldsymbol{\theta}_d$.
- Moreover,

$$\pi_d = \boldsymbol{\pi} \cdot \boldsymbol{\theta}_{\mathrm{d}} = \boldsymbol{\pi}^\top \boldsymbol{\theta}_{\mathrm{d}} = \boldsymbol{\psi}^\top \mathbf{D}\boldsymbol{\theta}_{\mathrm{d}} = \boldsymbol{\psi}^\top \mathbf{d} = \boldsymbol{\psi} \cdot \mathbf{d}$$

- Thus the no-arbitrage price of d is

$$\pi_d = \boldsymbol{\psi} \cdot \mathbf{d} = \sum_{s=1}^{S} \psi_s d_s.$$

FTAP and risk-neutral pricing I

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- We can rewrite the no-arbitrage pricing formula $\pi_d = \psi \cdot \mathbf{d}$ as an expectation.
- Suppose there is a riskfree asset with payoff 1 in every state. Its no-arbitrage price is $\sum_{s=1}^{S} \psi_s$, i.e.$\sum_{s=1}^{S} \psi_s = e^{-rT}$.
- Define the probability measure Q, with the probability of state s given by

$$q_s = \frac{\psi_s}{\sum_{s=1}^{S} \psi_s}.$$

- Since $\psi_s = e^{-rT} q_s$, we have

$$\pi_d = \sum_{s=1}^{S} \psi_s d_s = e^{-rT} \sum_{s=1}^{S} q_s d_s = e^{-rT} \mathbb{E}^Q(\tilde{d}),$$

where $\tilde{d}$ dis the payoff $\mathbf{d}$ written as a random variable.

FTAP and risk-neutral pricing II

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Assuming that there is a riskfree asset (or it can be replicated), we can restate the FTAP as follows:
  There is no arbitrage if and only if there exists a probability measure Q such that the price of any security that can be replicated by the underlying assets is equal to its expected payoff under Q, discounted at the riskfree rate.
- Q is called a "risk-neutral" probability measure.
- The FTAP is the cornerstone of derivatives pricing. The risk-neutral probabilities can be recovered from the underlying security prices and used to price redundant securities.

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FTAP and risk-neutral pricing III

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- In summary, any attainable payoff d can be priced by no-arbitrage:

$$\pi_d = \psi \cdot \mathbf{d} = e^{-rT}\mathbb{E}^Q(\tilde{d})$$

  which is equal to the price of any replicating portfolio.
- We say that markets are complete if any payoff $\mathbf{d} \in \mathbb{R}^S$ is attainable.
- If markets are complete, any payoff can be priced by no-arbitrage.

The static binomial model revisited

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- In the binomial model, the matrix

$$\begin{pmatrix} uS_0 & 1 \\ dS_0 & 1 \end{pmatrix}$$

  is nonsingular (since $u \neq d$).
- Therefore, markets are complete: Any security can be replicated by the stock and the riskfree asset.
- The state prices ψ_u and ψ_d solve the following system of equations:

$$\begin{pmatrix} S_0 & e^{-rT} \end{pmatrix} = \begin{pmatrix} \psi_u & \psi_d \end{pmatrix} \begin{pmatrix} uS_0 & 1 \\ dS_0 & 1 \end{pmatrix}$$

The static binomial model (cont'd)

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- The two state prices are

$$\psi_u = e^{-rT} q,$$
$$\psi_d = e^{-rT}(1-q),$$

where

$$q := \frac{e^{rT} - d}{u - d}$$

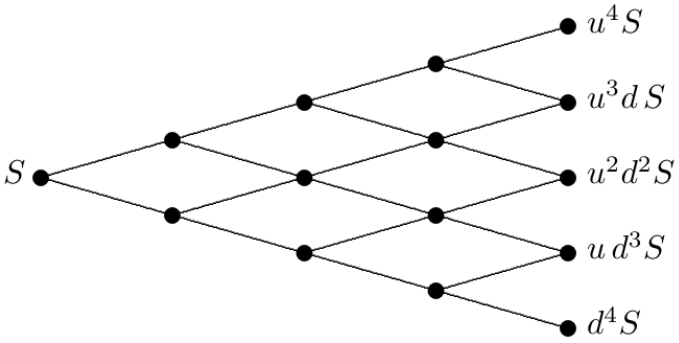
- There is no arbitrage iff $\psi \gg 0$ iff $0 < q < 1$iff $d < e^{rT} < u$.
- We can price any derivative by

$$V_0 = \psi_u V_u + \psi_d V_d = e^{-rT}[qV_u + (1-q)V_d].$$

Recombining binomial trees

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- The static binomial model extends straightforwardly to multiple periods.
- Consider an n-step binomial tree, with the same u,d and r parameters at each node.
- Then the evolution of the stock price can be described by a recombining binomial tree:



Recombining binomial trees (cont'd)

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Since in each one-step subtree, the u,d and r parameters are the same, so are the risk-neutral probabilities.
- As in the static case, any claim with payoff of the form $g(S_T)$ at time T can be priced either by replication or by risk-neutral valuation.
- We work backwards from the end of the tree.

Valuation in a recombining binomial tree

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

Replication:

- At each node at the penultimate date, choose the portfolio that replicates the payoff in the two succeeding nodes at the terminal date.
- At each node prior to this, choose the portfolio that finances the cost of the portfolio in the two succeeding nodes.
- The no-arbitrage price of the claim is the initial cost of this self-financing replicating strategy.

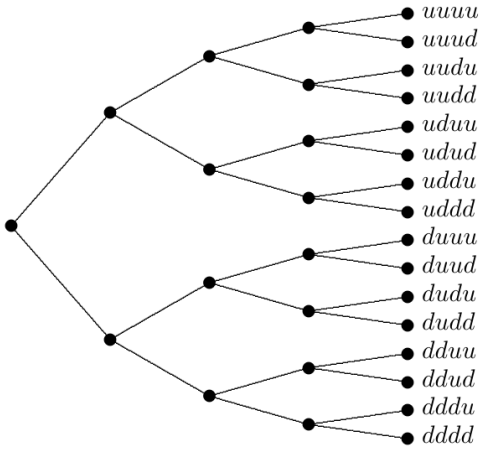
Risk-Neutral Valuation:

- Calculate the value of the claim at each node as the risk-neutral expectation of the value at the two succeeding nodes, discounted at the riskfree rate.
- Work backwards through the tree. The value at the initial node is the no-arbitrage price of the claim.

Recombining binomial trees (cont'd)

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

For path-dependent derivatives(e.g.barrier options), we have to use a non-recombining tree.



General multiperiod trees

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- If the u and d parameters are not the same in every one-step tree, the evolution of the stock price typically cannot be represented by a recombining tree.
  - The risk-neutral probabilities will typically be different for different subtrees.
- In a general multi-period tree, the number of branches may also differ across subtrees.
  - We can still price any attainable claim paying off at the end of the tree by working backwards through the tree.

Concluding remarks

Introduction
Binomial and FTAP
Binomial Model
Replicating portfolio
No-arbitrage pricing
Risk-neutral valuation
▷ FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- In the remainder of the course, we will cover a wide range of derivatives. Our goal will be to price and hedge those derivatives.
- The binomial model is the simplest possible model. We will consider models that allow for richer underlying dynamics.
- The insights of this chapter extend to these settings and we will use them over and over.
  - No-arbitrage arguments will be central.
- For a given product, the pricing and the hedging strategy (usually) depend on the model being used.
  - The specific assumptions of each model matter.

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Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

Appendix A

Outline

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

- Appendix A contains proofs of some results
- This material will not be examined

$\mathbb{E}[(X-K)^+]$ for X lognormal

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

- To derive the Black-Scholes formula, we used the following result: Suppose X is lognormally distributed, and $\text{Var}(\log X) = \sigma^2(T-t)$. Then

$$\mathbb{E}[(X-K)^+] = \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2),$$

where Φ is the standard normal cdf and

$$d_1 : \frac{\log[\mathbb{E}(X)/K] + \sigma^2(T-t)/2}{\sigma\sqrt{T-t}}, \quad d_2 := d_1 - \sigma\sqrt{T-t}$$

- To simplify notation, let $\mathbb{E}(\log X) = \alpha$, and $\text{Var}(\log X) = \beta^2$ (i.e. $\beta := \sigma\sqrt{T-t}$)

Proof

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

- We can write $\log X = \alpha + \beta z$, where z is standard normal. Thus $X = e^{\alpha+\beta z}$ and $\mathbb{E}(X) = e^{\alpha+\beta^2/2}$. Moreover, $X \geq K$ is equivalent to $\log X \geq \log K$,or

$$\begin{aligned} z &\geq \frac{\log K - \alpha}{\beta} \\ &= \frac{\log K - \log \mathbb{E}(X) + \beta^2/2}{\beta} \\ &= -\frac{\log[\mathbb{E}(X)/K] - \beta^2/2}{\beta} \\ &= -d_2. \end{aligned}$$

- Recall the standard normal density: $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$
- Let Φ be the standard normal cdf.

Proof(cont'd)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

Thus we have:

$$\begin{aligned}
\mathbb{E}[(X-K)^+] &= \int_{-d_2}^{\infty} (e^{\alpha+\beta z} - K)\phi(z)dz \\
&= \int_{-d_2}^{\infty} e^{\alpha+\beta z}\phi(z)dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \int_{-d_2}^{\infty} \mathbb{E}(X)e^{-\frac{1}{2}\beta^2+\beta z}\cdot\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2-\beta}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_1}^{\infty}\phi(z-\beta)d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2).
\end{aligned}$$

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Breeden-Litzenberger

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

- Let $\varphi(\cdot)$ be the density of S_T under the risk-neutral probability Q The price of a call option at $t = 0$ as a function of the strike price K can be written as:

$$C(K) = e^{-rT}\mathbb{E}^Q[(S_T - K)^+] = e^{-rT}\int_K^\infty (x - K)\varphi(x)dx$$

- Result [Breeden-Litzenberger (1978)]:

$$\varphi(K) = e^{rT}C''(K).$$

Proof of Breeden-Litzenberger result

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

Proof: We can write

$$e^{rT}C(K) = \int_K^\infty x\varphi(x)dx - K\int_K^\infty \varphi(x)dx.$$

Differentiating:

$$\begin{aligned} e^{rT}C'(K) &= \frac{d}{dK}\int_K^\infty x\varphi(x)dx - K\frac{d}{dK}\int_K^\infty \varphi(x)dx - \int_K^\infty \varphi \\ &= -K\varphi(K) - K[-\varphi(K)] - \int_K^\infty \varphi(x)dx \\ &= -\int_K^\infty \varphi(x)dx. \end{aligned}$$

Differentiating again:

$$e^{rT}C''(K) = -[-\varphi(K)] = \varphi(K).$$

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
▷ Appendix B

USEFUL FACTS

USEFUL FACTS

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
▷ Appendix B

- Put-call parity (non-dividend paying stock):

$$C_t = P_t + S_t - e^{-r(T-t)}K$$

- Lognormal property of GBM:
  Suppose $dS_t = \mu S_t dt + \sigma S_t dW_t$. Then the conditional distribution of $\log S_T$ at time t is normal:

$$\log S_T \sim \mathcal{N}\left[\log S_t + \left(\mu - \sigma^2/2\right)(T-t), \sigma^2(T-t)\right]$$

- Black-Scholes formula:
  The price of a European call option at $t < T$ is given by

$$C(S,t) = S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2)$$

  where $\Phi(\cdot)$ is the standard normal cdf, and

$$d_1 = \frac{\log(S/K) + \left(r + \sigma^2/2\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

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USEFUL FACTS

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
▷ Appendix B

- Black-Scholes PDE:
  Consider a derivative on a stock with payoff $g(S_T)$ at time T, whose price at time $t \leq T$ is given by $F(S,t)$. The Black-Scholes PDE is

$$F_t(S,t) + rSF_S(S,t) + \frac{1}{2}\sigma^2 S^2 F_{SS}(S,t) - rF(S,t) = 0$$

- Standard normal density:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- Expectation of the exponential of a normal random variable:
  If X normally distributed, then

$$\mathbb{E}[\exp(X)] = \exp\left[\mathbb{E}(X) + \frac{1}{2}\operatorname{Var}(X)\right]$$

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